Blockchain Smart Contract Transaction Anomaly Detection Project Report

Ayça Görgülü - 22290494

Duru Yıldırım - 23291391

Mina Özdoğan - 23291224

Project Objective

The objective of this project is to develop and evaluate an unsupervised machine learning framework for identifying anomalous smart contract transactions in decentralized finance (DeFi) environments. As blockchain-based financial protocols increasingly facilitate high-volume activities such as token trading and liquidity provision, detecting abnormal transaction behavior has become critical for improving transparency, trust, and risk awareness in on-chain financial systems.

Using real-world transaction data from the Uniswap V3 USDC/WETH liquidity pool, collected via the Etherscan API, this study analyzes transaction-level and execution-related features derived from decoded Ethereum event logs. The project applies unsupervised anomaly detection techniques—specifically Isolation Forest and Autoencoder neural networks—to model normal smart contract behavior and identify transactions that deviate significantly from learned patterns, without relying on labeled examples of fraud or attacks.

By combining anomaly signals from multiple models into a unified risk assessment framework, the project aims to highlight high-confidence anomalous transactions that warrant further investigation. Rather than labeling transactions as fraudulent, the analysis focuses on identifying statistically and behaviorally unusual activity, such as extreme transaction sizes, repeated address interactions, and structured temporal patterns, which may indicate automated or exploitative behavior.

Overall, this project seeks to demonstrate how unsupervised machine learning methods, supported by interpretable visual and behavioral analyses, can enhance monitoring and risk screening in decentralized financial ecosystems. The proposed approach is fully data-driven, adaptable to other smart contracts or protocols, and contributes to ongoing efforts to improve transparency and operational oversight in blockchain-based finance.

Dataset Description

The dataset used in this study is titled "Uniswap V3 Ethereum Transaction Log Dataset" and was constructed specifically for this project using data obtained through the Etherscan API, a widely used public interface for retrieving blockchain transaction and log information. The raw dataset consists of Ethereum event logs associated with the Uniswap V3 USDC/WETH liquidity pool. These raw logs include the standard Ethereum log fields—address, topics, data, blockNumber, blockHash, timeStamp, gasPrice, gasUsed, logIndex, transactionHash, and transactionIndex. A total of N = 1000 raw observations were retrieved. These fields represent a mixture of categorical, numerical, textual, and date-related variables. Although the raw dataset provides the fundamental blockchain metadata, it is not directly suitable for anomaly detection because many of its fields are stored as hexadecimal values or encoded structures.

To transform the raw logs into a structured and analyzable format, a comprehensive preprocessing workflow was applied. First, hexadecimal numerical values (specifically blockNumber and timeStamp) were converted into integer format, enabling chronological sorting and time-based analysis. The Unix timestamps were further transformed into a human-readable datetime format (timeStamp_dt) to facilitate temporal interpretation. A critical stage of preprocessing involved decoding Uniswap V3 swap-specific information from the topics and data fields. Since Ethereum logs store event parameters in a 256-bit encoded format, custom decoding functions were implemented to extract meaningful transaction attributes. Using the topics array, the sender and recipient wallet addresses were derived by parsing the 32-byte indexed fields. The non-indexed data field was split into five 32-byte segments corresponding to the core Uniswap swap parameters: amount0, amount1, sqrtPriceX96, liquidity, and tick. Signed decoding logic was applied to amount0, amount1, and tick to correctly interpret negative values, representing the direction of the swap.

After preprocessing, the final processed dataset contains k = 21 variables, representing both the raw blockchain metadata and the decoded financial and behavioral attributes required for anomaly detection.

The variables can be classified into four categories:

(1) Blockchain metadata: address (categorical, emitting contract address), blockNumber (hex/string), blockNumber_int (numeric, integer block index), blockHash (text), logIndex (numeric), transactionIndex (numeric), and transactionHash (categorical), which provide structural details about each transaction's position on the chain.

(2) Temporal variables: timeStamp (hex/string), timeStamp_int (numeric, seconds since epoch), and timeStamp_dt (datetime), which describe when transactions occurred.

(3) Execution-related variables: gasPrice (numeric, measured in wei) and gasUsed (numeric, gas units), both of which are important indicators of transaction cost and computational complexity.

(4) Swap-specific financial variables: sender (categorical), recipient (categorical), amount0 and amount1 (numeric, raw token amounts), sqrtPriceX96 (numeric, fixed-point price representation), liquidity (numeric, pool liquidity level), and tick (numeric, integer tick value representing the price level). These decoded variables contain the core economic signals necessary to detect abnormal transaction behaviors, unusual trade amounts, or liquidity manipulation within the pool.

Overall, the processed dataset provides a rich combination of numerical, categorical, and temporal variables that describe both smart-contract execution patterns and economic characteristics of Uniswap V3 swaps. While the raw Etherscan data provides the foundational blockchain log fields, it is the processed dataset—generated through extensive decoding, conversion, and augmentation—that is ultimately used for statistical analysis and unsupervised anomaly detection. This structured dataset forms the analytical basis for the machine learning models applied in the later sections of the project.

Frequency Analysis of Sender and Recipient Addresses

To understand user behavior in the Uniswap V3 pool better, how frequently each wallet address appears as a sender or recipient in the swap logs is examined. This helps reveal whether the activity is spread across many users or dominated by a small number of highly active wallets (e.g., trading bots, market makers, or arbitrageurs).

The system counts how many times each address initiates a swap transaction as a sender and how many times each address receives the output tokens as a recipient. After aggregating this data, it identifies and displays the top 10 most active addresses for each category. These frequency tables are then used as the basis for creating bar charts and pie charts that visualize the distribution of swap activity across different addresses in the network. Analyzing the most active addresses can help identify repetitive behavior, automated trading patterns, and potential sources of anomalies.

```
(sender
 0x3fc91a3afd70395cd496c647d5a6cc9d4b2b7fad    514
 0xef1c6e67703c7bd7107eed8303fbe6ec2554bf6b     86
 0xe592427a0aece92de3edee1f18e0157c05861564     69
 0x1111111254eeb25477b68fb85ed929f73a960582     57
 0xdef1c0ded9bec7f1a1670819833240f027b25eff     53
 0x68b3465833fb72a70ecdf485e0e4c7bd8665fc45     47
 0x3208684f96458c540eb08f6f01b9e9afb2b7d4f0     28
 0xe8cfad4c75a5e1caf939fd80afcf837dde340a69     17
 0x51c72848c68a965f66fa7a88855f9f7784502a7f     15
 0xd7f3fbe8c72a961a5515203eada59750437fa762     11
 Name: count, dtype: int64,
 recipient
 0x3fc91a3afd70395cd496c647d5a6cc9d4b2b7fad    287
 0xef1c6e67703c7bd7107eed8303fbe6ec2554bf6b     83
 0xdef1c0ded9bec7f1a1670819833240f027b25eff     36
 0x1111111254eeb25477b68fb85ed929f73a960582     33
 0xe592427a0aece92de3edee1f18e0157c05861564     25
 0x81a460ea6fd96a73d5672f1f4aa684697d4b44cc     21
 0xe8cfad4c75a5e1caf939fd80afcf837dde340a69     17
 0x3208684f96458c540eb08f6f01b9e9afb2b7d4f0     16
 0x51c72848c68a965f66fa7a88855f9f7784502a7f     15
 0xdef171fe48cf0115b1d80b88dc8eab59176fee57     14
 Name: count, dtype: int64)
```

Interpretation of Descriptive Statistics

	count	mean	std	min	25%	50%	75%	max	skewness	kurtosis
amount0	1000.0	-3.599611e+09	4.974180e+10	-5.239314e+11	-1.474226e+09	85419316.5	1.301729e+09	3.231795e+11	-3.269262	29.861463
tick	1000.0	2.022587e+05	1.003190e+01	2.022340e+05	2.022510e+05	202263.0	2.022660e+05	2.022710e+05	-0.653976	-0.902608

The table summarizes the key descriptive statistics for the numerical variables used in the analysis, including trade amounts, price-related indicators, liquidity, and tick levels. The variables show substantial variation in scale and distribution, reflecting the heterogeneous nature of swap activity on the Uniswap V3 USDC/WETH pool.

For amount0, the mean and median differ significantly, and the standard deviation is extremely large relative to the mean. This pattern indicates a highly dispersed distribution with extreme values present in both positive and negative directions. The minimum and maximum values demonstrate that while many swaps involve small amounts, a small number of transactions involve very large amounts of USDC, contributing to the wide spread. The strongly negative skewness value shows that the distribution is left-skewed, meaning that large negative trades (large amounts paid into the pool) occur more frequently than large positive trades.

The very high kurtosis value confirms a heavy-tailed distribution, which is typical for decentralized exchange data where a small number of swaps dominate transaction volume.

For tick, the statistics reveal that tick values remain within a relatively narrow interval, reflecting price stability within the observed period. The mean and median values are close, and the standard deviation is small compared to the overall range, indicating a concentrated distribution. Tick skewness is slightly negative, suggesting a mild tilt toward lower tick values, while kurtosis is slightly negative, indicating a distribution that is flatter than a normal distribution. These results are consistent with the continuous and liquid nature of the USDC/WETH market, where prices tend to fluctuate smoothly without extreme jumps.

Together, these descriptive statistics highlight the contrast between the extreme variability of swap sizes and the relative stability of price-related variables. This discrepancy supports the use of anomaly detection methods, as unusual transaction sizes or activity patterns can be meaningfully distinguished from typical behavior.

Data Quality Control

A data quality assessment was performed on the processed Uniswap V3 USDC–WETH swap dataset. Missing value counts were computed for all variables, and the results show that both the raw metadata fields obtained from Etherscan and the decoded numerical fields contain zero missing values. This indicates that all 1000 retrieved log entries were complete and fully usable, so no rows required removal or imputation.

Outlier analysis was conducted using the Interquartile Range (IQR) method for the numerical variables amount0, amount1, sqrtPriceX96, liquidity, and tick. The results indicate the presence of 230 outliers in amount0, 230 in amount1, and 455 in liquidity, while sqrtPriceX96 and tick showed no outliers. These large deviations, particularly in swap amounts and liquidity, reflect real market behavior on decentralized exchanges, where high-volume transactions and rapid liquidity adjustments are common. Because such extreme values may correspond to meaningful events—including arbitrage, liquidations, or unusual trading activity—these outliers were not removed and were retained for later anomaly detection analysis.

Since the dataset contained complete entries and the outliers were meaningful for modeling purposes, no missing data handling or outlier removal methods (such as winsorization or trimming) were applied. All data points were preserved to maintain the integrity and variability of real-world DeFi transaction behavior.

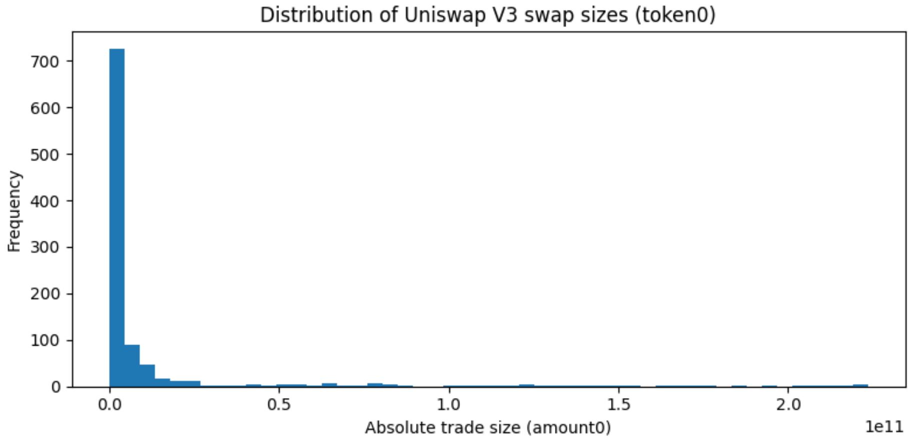
```
{'amount0': 230,
 'amount1': 230,
 'sqrtPriceX96': 0,
 'liquidity': 455,
 'tick': 0}
```

Graphical Analysis

A series of six visualizations was generated to examine the structure and behavior of swap activity within the Uniswap V3 pool. Each plot highlights a different aspect of the dataset and contributes to understanding transaction patterns prior to anomaly detection.

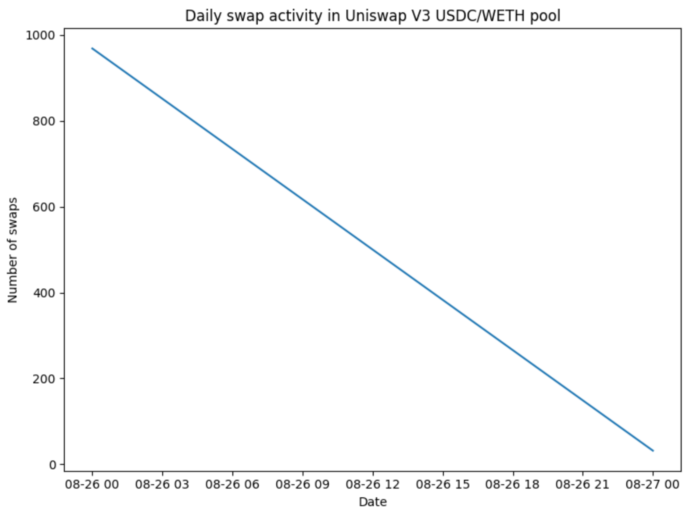
1. Histogram of Swap Sizes (amount0)

The histogram illustrates the distribution of absolute trade sizes on the USDC (token0) side. The distribution is highly right-skewed, indicating that most swaps involve relatively small amounts, while a limited number of transactions exhibit significantly larger sizes. Such extreme values are often associated with whale trades, arbitrage operations, or automated strategies.



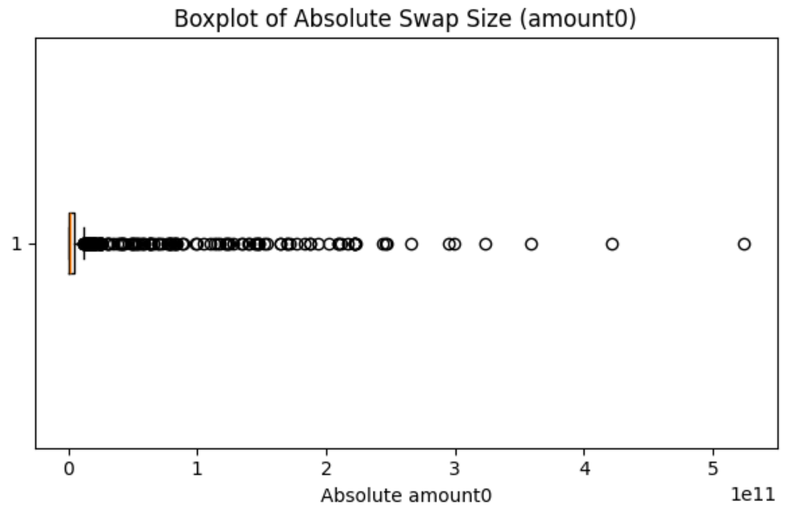
2. Time Series of Daily Swap Activity

The time series plot shows the variation in swap counts across days within the examined block range. Activity levels fluctuate over time, reflecting changes in market conditions, token price dynamics, or user engagement. Sudden increases or decreases in swap frequency may indicate periods of heightened volatility or unusual behavior.



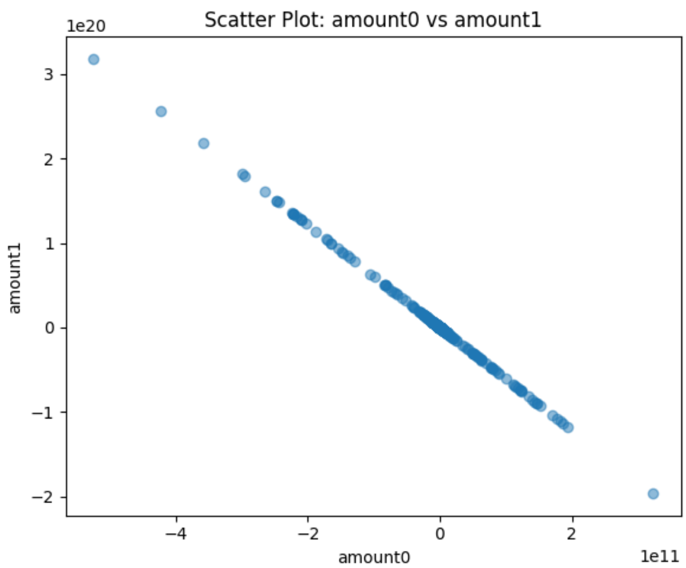
3. Boxplot of Absolute Swap Sizes

The boxplot further emphasizes the skewed distribution of swap sizes. The central quartiles are compressed near zero, while numerous outliers extend far to the right. These outlier observations represent unusually large trades and are relevant for subsequent anomaly detection, as they deviate from typical transactional behavior.



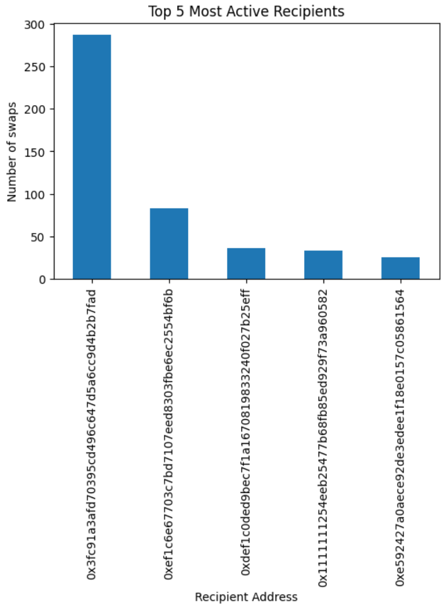
4. Scatter Plot of amount0 vs amount1

The scatter plot visualizes the relationship between the USDC (amount0) and WETH (amount1) quantities exchanged in each swap. The diagonal pattern reflects the pool's price dynamics, with most points following a consistent ratio. A small number of points deviate from this trend, suggesting possible irregular trades or atypical price movements.



5. Bar Chart of the Top 5 Recipient

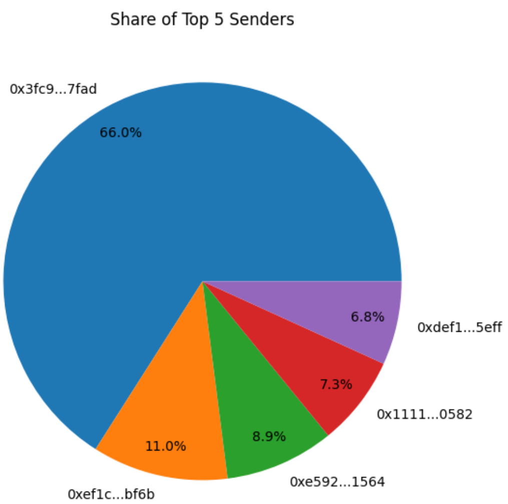
The bar chart displays the five most active recipient addresses within the dataset. A notable concentration of activity is observable among a small subset of wallets, consistent with the presence of high-frequency trading bots, market makers, or automated liquidity strategies.



6. Pie Chart of the Top 5 Senders' Share

The pie chart provides a proportional view of swap activity attributed to the top five senders. The distribution confirms that a limited number of addresses account for a substantial share of overall transactions, highlighting the dominance of a few participants within the pool.

Overall, these visualizations reveal key structural characteristics of the dataset, including skewed trade sizes, concentrated participation, and temporal variability. This descriptive analysis establishes a foundation for applying anomaly detection methods.



Numerical Feature Preparation and Robust Scaling

Prior to anomaly detection, numerical transaction attributes are prepared and standardized to ensure consistency and reliability in downstream analysis. A working copy of the cleaned transaction dataset is first created to preserve the original data and support reproducible transformations. The numerical feature set consists of transaction-level and execution-related attributes, including transferred token amounts, liquidity measures, pricing variables, and gas-related metrics. Since certain blockchain data fields may be represented in hexadecimal format, a conversion function is applied to transform hexadecimal strings into their corresponding numerical values while leaving already numeric entries unchanged. This step ensures that all selected features are represented in a uniform numerical format suitable for statistical modeling. After conversion, the numerical features are assembled into a feature matrix and scaled using Robust Scaling. RobustScaler normalizes each feature based on its median and interquartile range rather than the mean and standard deviation. This approach is particularly appropriate for blockchain transaction data, which often exhibits heavy-tailed distributions and extreme values. By reducing the influence of outliers, robust scaling improves model stability and prevents extreme observations from disproportionately affecting the anomaly detection process. The scaled feature values are then re-integrated into the dataset as additional columns, preserving the original feature values alongside their standardized counterparts. This augmented dataset provides a consistent input representation for subsequent anomaly detection models while retaining interpretability and flexibility for analysis and visualization.

Application of the Isolation Forest Model

Isolation Forest is applied as an unsupervised anomaly detection method to identify atypical smart contract transactions based on their numerical characteristics. Prior to model training, all relevant numerical features are converted into a consistent numerical format and scaled to ensure comparability across variables with different magnitudes. The resulting scaled feature matrix is preserved and reused as a standardized model input, ensuring consistency and reproducibility across analytical stages.

The model is trained on the full scaled transaction feature set using an ensemble of isolation trees. A fixed contamination ratio is specified to reflect the expected proportion of anomalous transactions and to determine the internal decision threshold. Upon training, the model assigns each transaction both a binary anomaly label, indicating normal or anomalous behavior, and a continuous anomaly score that quantifies the degree of deviation from typical transaction patterns.

Following model inference, anomaly labels and scores are merged back with the processed transaction records to preserve transaction-level context, including identifiers, addresses, and timestamps. Transactions are then partitioned into normal and anomalous subsets to support targeted analysis. The continuous anomaly scores enable ranking of transactions by anomaly severity, while the binary labels facilitate aggregation, visualization, and comparative feature analysis.

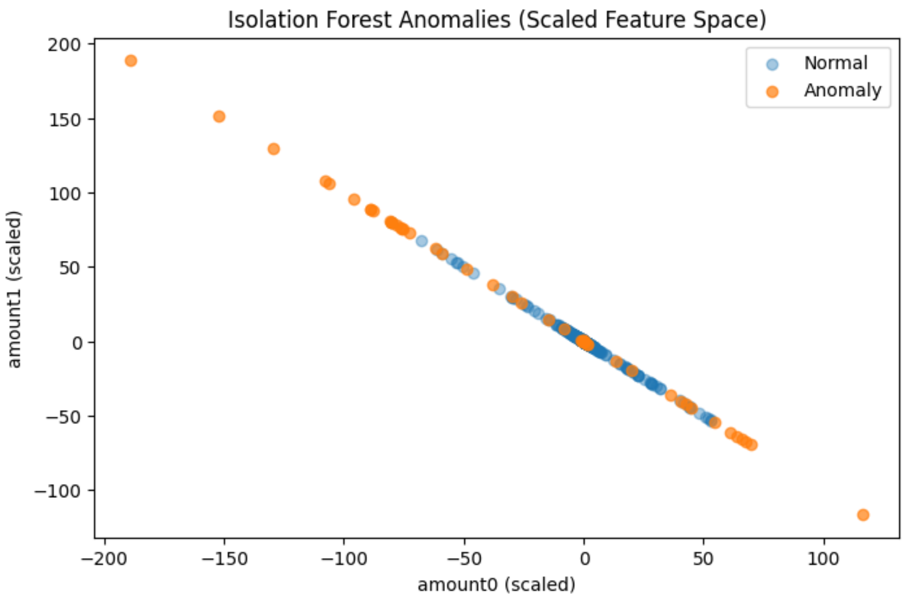
Isolation Forest Results and Observational Insights

The application of the Isolation Forest model provides a structured, unsupervised characterization of anomalous behavior within the smart contract transaction dataset. Based on the specified contamination level, the model identifies a distinct subset of transactions as anomalous, while the majority are classified as normal. This partitioning establishes a clear distinction between typical transaction behavior and transactions exhibiting statistically unusual patterns.

Analysis of the Isolation Forest anomaly scores indicates that transactions with the highest anomaly risk are frequently associated with extreme values in key numerical features, particularly large token transfer amounts and abrupt changes in liquidity-related metrics. These findings suggest that the model effectively captures deviations driven by unusually large-scale or disruptive transactional activity.

Further examination of anomalous transactions reveals non-uniform participation across sender and recipient addresses. Certain wallets appear repeatedly within the anomalous subset, indicating concentrated abnormal behavior rather than uniformly distributed noise. This concentration suggests that anomalous activity may be linked to specific entities or automated behaviors, rather than random fluctuations in transaction data.

Feature-level comparisons between normal and anomalous transactions highlight systematic deviations across multiple transaction attributes. The relative magnitude of these deviations provides insight into which swap characteristics contribute most strongly to anomaly detection, supporting interpretability of the model outputs. Collectively, these findings demonstrate that the Isolation Forest model not only identifies anomalous transactions but also enables meaningful interpretation of the behavioral patterns underlying these anomalies.



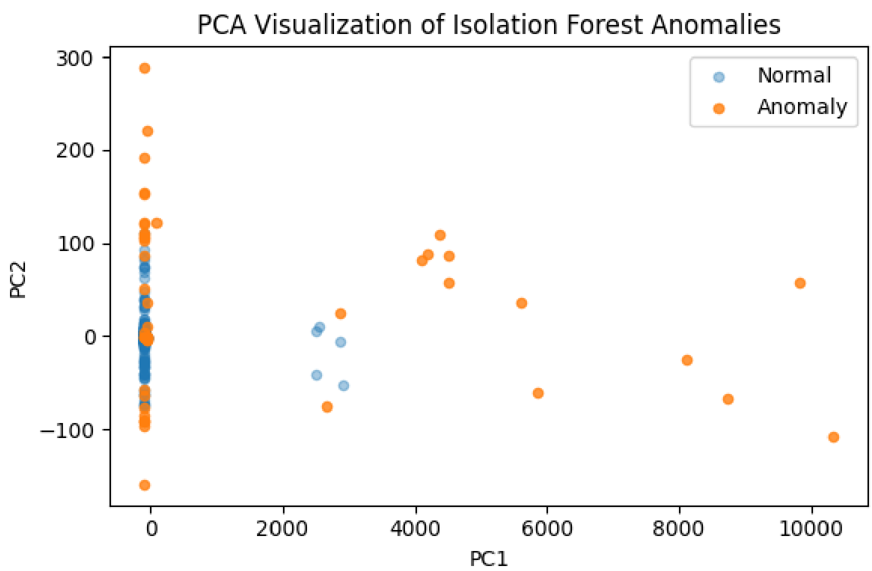
PCA-Based Visualization of Isolation Forest Anomalies

To support interpretability of the Isolation Forest results, Principal Component Analysis (PCA) is applied as a post-hoc dimensionality reduction technique to the scaled numerical feature matrix. PCA is used exclusively for visualization and does not influence the anomaly detection process. Its purpose is to provide a low-dimensional representation of transaction behavior that facilitates qualitative inspection of anomaly separation.

In this analysis, the first two principal components are extracted, forming a two-dimensional embedding of the transaction feature space. The resulting PCA coordinates are aligned with the original transaction indices and augmented with the Isolation Forest anomaly labels to preserve the association between each transaction and its classification.

A scatter plot of the two principal components visually distinguishes normal and anomalous transactions. Normal transactions form dense clusters within the reduced feature space, reflecting consistent and regular smart contract activity. In contrast, anomalous transactions appear more dispersed and frequently occupy regions farther from the main concentration of normal observations, particularly along the dominant variance direction captured by the first principal component.

Although PCA is not designed for anomaly detection, the observed dispersion and partial separation of anomalous transactions in the reduced feature space provide qualitative evidence that the Isolation Forest is identifying transactions with structurally distinct characteristics. This visualization therefore enhances interpretability and supports confidence in the detected anomalies by illustrating how abnormal behavior manifests across the dominant variance directions of the data.



Application of the Autoencoder Model

An autoencoder is applied as a reconstruction-based anomaly detection method to complement the Isolation Forest analysis. The model operates on the same scaled numerical feature representation used previously, ensuring consistency across anomaly detection approaches. To avoid learning anomalous patterns, the autoencoder is trained exclusively on transactions classified as normal by the Isolation Forest model.

The autoencoder architecture follows a symmetric, fully connected design consisting of an encoder that compresses the input feature space into a low-dimensional bottleneck representation and a decoder that reconstructs the original input from this compressed form. Nonlinear activation functions are employed in the hidden layers to capture complex feature interactions, while a linear output layer supports accurate reconstruction of scaled numerical values. The model is optimized using mean squared error loss, which directly measures reconstruction fidelity.

During training, the normal transaction subset is divided into training and validation sets to monitor reconstruction performance on unseen normal data and to ensure generalizable learning. After training, the autoencoder is applied to reconstruct the feature vectors of all transactions. For each transaction, a reconstruction error is computed as the mean squared difference between the original and reconstructed feature vectors. This reconstruction error serves as a continuous anomaly score, reflecting the degree of deviation from learned normal transaction behavior.

To obtain discrete anomaly labels, a percentile-based thresholding strategy is applied to the reconstruction error distribution. Transactions with reconstruction errors exceeding the selected threshold are classified as anomalous, while the remaining transactions are classified as normal. Both the continuous reconstruction error and the resulting anomaly labels are stored alongside transaction metadata to support downstream analysis and integration with other anomaly detection outputs.

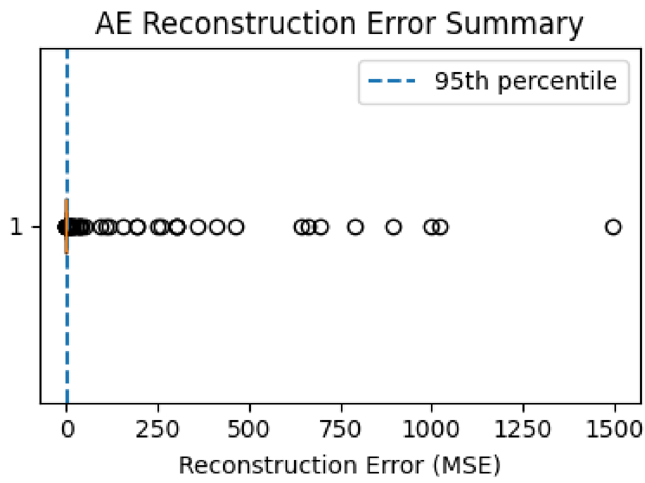
Autoencoder Results and Observational Insights

The autoencoder-based analysis reveals that the majority of transactions are reconstructed with low error, indicating strong conformity to the learned representation of normal transaction behavior. A smaller subset of transactions exhibits substantially higher reconstruction errors, reflecting structural deviations from typical patterns captured during training.

The distribution of reconstruction errors is characterized by a pronounced right tail, consistent with the presence of atypical transaction behavior. Visualization of the reconstruction error distribution, together with the applied threshold, illustrates a clear separation between typical reconstruction behavior and extreme deviations. This separation supports the suitability of reconstruction error as an anomaly signal within the transaction dataset.

Transactions classified as anomalous by the autoencoder represent observations that differ significantly from normal behavior in the scaled feature space, rather than random noise. The percentile-based threshold provides a transparent and reproducible criterion for anomaly identification, enabling consistent labeling across the dataset.

Overall, the autoencoder contributes a reconstruction-driven perspective on anomalous transaction behavior, capturing deviations that arise from the internal structure of normal transaction patterns. These results establish a complementary anomaly signal that is later combined with Isolation Forest outputs to support comparative analysis and unified risk assessment.



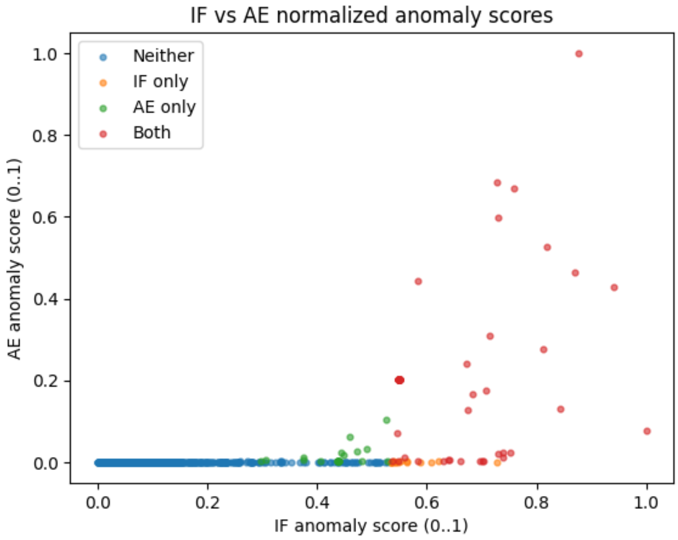
Comparative Anomaly Analysis and High-Confidence Risk Assessment

Anomaly Score Comparison and Model Agreement

After obtaining anomaly scores from the Isolation Forest and Autoencoder models, a comparative analysis is performed to evaluate agreement between the two unsupervised detection approaches. Each transaction is associated with two continuous anomaly signals: the Isolation Forest score, which reflects how easily a transaction is isolated within the feature space, and the Autoencoder reconstruction error, which captures deviation from learned normal transaction patterns.

The anomaly score comparison figure generated in the notebook visualizes the relationship between these two scores. Most transactions cluster in regions corresponding to low anomaly scores for both models, indicating consistent normal behavior. In contrast, a smaller subset of transactions exhibits elevated scores in one or both dimensions, suggesting abnormality under at least one detection mechanism.

Transactions flagged as anomalous by both models form a distinct region in the comparison plot. These overlapping detections are treated as high-confidence anomalies, as agreement between two fundamentally different modeling approaches provides stronger evidence of genuine deviation from normal smart contract behavior.



High-Confidence Anomalies and Fusion-Based Risk Stratification

To improve robustness and reduce model-specific bias, anomaly signals from both models are combined into a unified fusion score. This score aggregates information from Isolation Forest and Autoencoder outputs, enabling transactions to be ranked by overall abnormality rather than relying on a single detection criterion.

Based on the fusion score distribution, transactions are grouped into interpretable risk tiers, including low, medium, and high risk. The results show that the majority of transactions fall within the low-risk category, reflecting stable and routine smart contract interactions. A relatively small subset is classified as high risk, corresponding primarily to transactions flagged consistently by both models.

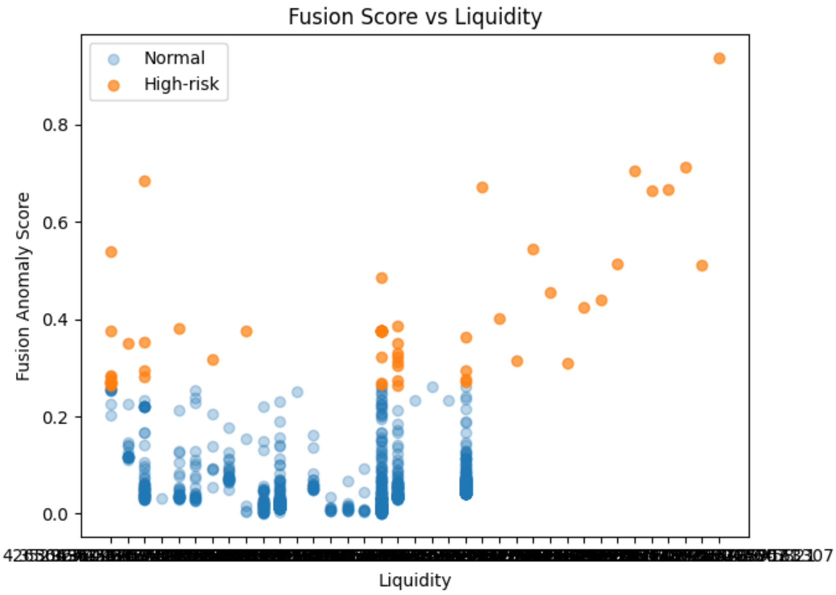
This fusion-based stratification provides a practical mechanism for prioritizing transactions for further investigation while maintaining an unsupervised and data-driven detection framework.

Behavioral Characteristics of High-Risk Transactions

Liquidity-Related Patterns

The relationship between fusion score and liquidity is examined using the liquidity-focused plot generated in the notebook. High-risk transactions are observed across a wide range of liquidity values, including periods of deep liquidity. This pattern is consistent with Uniswap V3's concentrated liquidity design, where large liquidity values may exist within narrow price ranges.

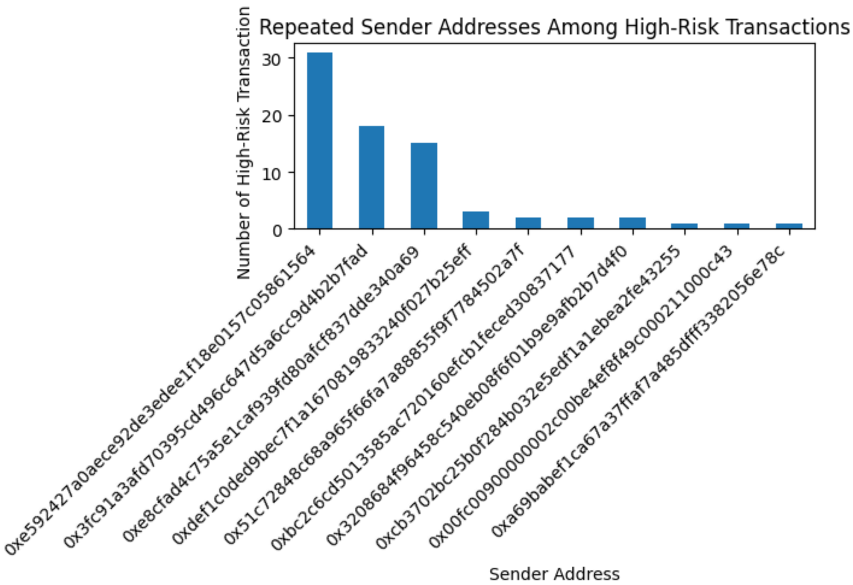
The presence of elevated fusion scores even under stable liquidity conditions suggests that anomalous behavior is not primarily driven by liquidity scarcity. Instead, these anomalies are more likely associated with atypical transaction execution patterns or strategic behavior, such as automated interactions with the pool.



Repeated Sender Address Behavior

Address-level analysis of high-risk transactions reveals a strong concentration among a small number of sender addresses. The bar chart produced in the notebook shows that a limited set of wallets accounts for a disproportionately large share of high-risk transactions.

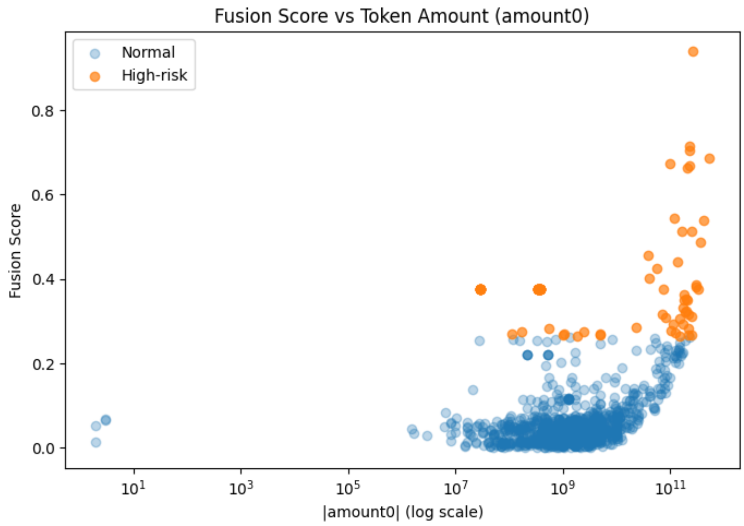
This repetition strongly indicates automated or algorithmic activity rather than sporadic human-driven behavior. The consistency of sender concentration across multiple high-risk transactions reinforces the reliability of the anomaly detection framework and supports the interpretation that detected anomalies reflect systematic behavioral patterns.



Token Amount Extremes

The scatter plot comparing fusion scores with token amounts (on a logarithmic scale) highlights a clear relationship between transaction size and anomaly severity. High-risk transactions are predominantly concentrated among observations involving exceptionally large token transfers.

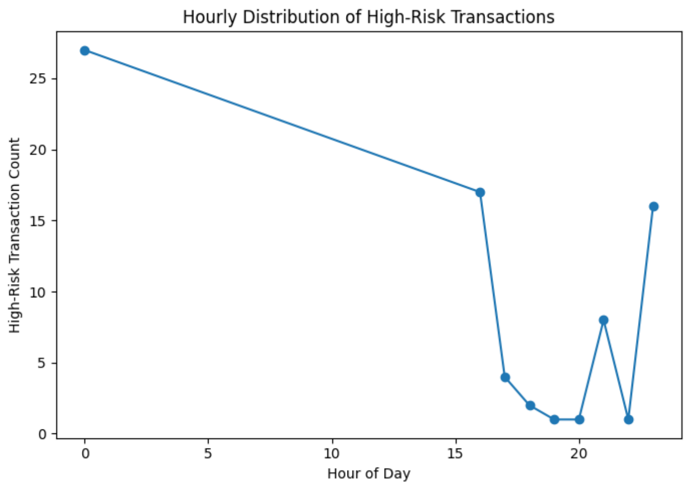
This pattern indicates that extreme transaction sizes are a major driver of anomalous behavior in the dataset. Such transactions may correspond to large-scale arbitrage, aggressive trading strategies, or other high-impact interactions that deviate significantly from typical swap behavior.



Temporal Regularity of High-Risk Transactions

Temporal analysis of anomalous activity is conducted using the hourly distribution plot generated in the notebook. The figure reveals that high-risk transactions do not occur uniformly across time. Instead, activity peaks at specific hours, forming a structured and recurring pattern.

This predictable temporal distribution provides further evidence of automation, as algorithmic systems often operate on fixed schedules. When combined with repeated sender address behavior, the temporal regularity strongly suggests that a substantial portion of high-risk anomalies is driven by programmed or bot-based execution rather than random market events.



Contribution to the Literature

This project contributes to the growing literature on blockchain analytics and decentralized finance by demonstrating how unsupervised machine learning techniques can be applied to real-world smart contract transaction data to identify anomalous behavior without relying on labeled examples. Unlike many existing studies that focus on price prediction, market efficiency, or supervised fraud classification, this work emphasizes anomaly detection as a risk-aware screening tool for on-chain financial activity.

A key contribution of this study is the construction and analysis of a custom transaction-level dataset derived from decoded Uniswap V3 Ethereum event logs. By transforming raw blockchain data into economically meaningful execution and liquidity-related features, the project provides a practical example of how low-level smart contract logs can be converted into structured inputs suitable for machine learning analysis. This dataset and preprocessing approach can serve as a reference for future research on decentralized exchange behavior.

Methodologically, the project contributes by combining two complementary unsupervised models—Isolation Forest and Autoencoder neural networks—and integrating their outputs into a unified fusion-based risk assessment framework. While both models are well-established individually, their joint application and agreement-based anomaly identification provide a robust and interpretable approach for reducing false positives in unsupervised settings. The inclusion of post-hoc PCA visualization and behavioral pattern analysis further enhances interpretability, which remains a key challenge in anomaly detection research.

The findings of this study are relevant to multiple stakeholders within the DeFi ecosystem. Protocol developers and startups may use similar frameworks to monitor abnormal interaction patterns and improve system resilience. Regulators and compliance analysts can benefit from unsupervised screening tools that flag statistically unusual activity without requiring predefined fraud labels. Investors and risk managers may also gain insights into automated or high-impact trading behaviors that influence liquidity and market stability. Overall, this project illustrates how data-driven anomaly detection can support transparency and risk awareness in blockchain-based financial systems.

Did the Project Meet Expectations?

The primary objective of this project was to develop an unsupervised framework capable of identifying anomalous smart contract transactions in a decentralized finance environment using real-world blockchain data. Based on the results obtained, this objective was successfully met. The applied models were able to distinguish normal transaction behavior from structurally and statistically unusual patterns, and the combined fusion approach enabled the identification of high-confidence anomalies for further analysis.

Several limitations were encountered during the project. First, the analysis is constrained to a single liquidity pool and a limited number of transactions, which may restrict the generalizability of the findings across different protocols or market conditions.

Second, as with all unsupervised approaches, the models identify deviations from learned norms rather than verified malicious or fraudulent behavior. As a result, detected anomalies require contextual interpretation and cannot be automatically classified as harmful activity.

From a modeling perspective, the performance of the anomaly detection methods depends on feature representation and threshold selection. While robust scaling and percentile-based thresholds were chosen to handle heavy-tailed distributions, alternative configurations or additional features could influence the sensitivity of the results. Despite these limitations, the consistency between Isolation Forest and Autoencoder outputs, supported by behavioral and temporal analyses, suggests that the detected patterns are meaningful rather than random artifacts.

Overall, the project met its intended goals by delivering a coherent, interpretable, and technically sound anomaly detection pipeline. The findings align with expectations regarding the presence of automated and high-impact transaction behavior in decentralized exchanges, and the framework provides a solid foundation for future extensions and applied research in blockchain-based financial analytics.

Possible Future Improvements

Future extensions of this work may include applying the proposed framework to longer time horizons or larger datasets to assess the stability and generalizability of the detected anomaly patterns across different market conditions. Incorporating graph-based interaction features between addresses, such as transaction networks or sender–recipient connectivity measures, could further enhance the detection of coordinated or collusive behavior that may not be fully captured by transaction-level features alone. Additionally, integrating protocol-specific domain knowledge—such as pool configuration parameters, fee tiers, or on-chain governance events—may improve interpretability and provide richer contextual explanations for detected anomalies. These enhancements could strengthen the operational relevance of the framework while preserving its unsupervised, data-driven nature.

Conclusion and Implications

This project presents a comprehensive, unsupervised framework for detecting anomalous smart contract transactions in decentralized finance environments, using real-world blockchain data from the Uniswap V3 USDC/WETH liquidity pool. Through systematic data collection, preprocessing, feature extraction, and model-based analysis, the study demonstrates how machine learning techniques can enhance transparency and risk awareness in high-volume, on-chain financial systems.

A detailed preprocessing pipeline transforms raw Ethereum event logs into a structured and analyzable dataset by decoding swap-specific parameters, converting hexadecimal fields into numerical representations, and constructing meaningful execution-related and financial features.

Exploratory and descriptive analyses reveal that while most transactions exhibit stable and predictable behavior, key variables such as swap amounts and liquidity display heavy-tailed distributions and extreme values. These characteristics motivate the use of robust scaling techniques and unsupervised anomaly detection methods capable of handling non-Gaussian data and meaningful outliers.

Isolation Forest is applied as the primary anomaly detection model to identify transactions that deviate from typical behavior based on numerical characteristics. The model successfully separates the dataset into normal and anomalous subsets, with high-risk transactions frequently associated with unusually large token transfers, abrupt liquidity changes, and concentrated address-level activity. Post-hoc PCA visualization further supports these findings by illustrating that anomalous transactions occupy structurally distinct regions in the reduced feature space, reinforcing the interpretability of the detection results.

To complement the Isolation Forest analysis, an Autoencoder neural network is trained exclusively on transactions classified as normal, enabling reconstruction-based anomaly detection. Transactions with high reconstruction error are identified as structurally dissimilar to learned normal patterns.

The Autoencoder captures deviations arising from complex feature interactions that may not be fully detected by tree-based methods, providing a complementary anomaly signal.

The outputs of both models are combined into a unified fusion score, enabling comparative analysis and high-confidence anomaly identification. Transactions consistently flagged by both models are classified as high-risk, significantly reducing false positives while preserving sensitivity to meaningful deviations. Behavioral analysis of these high-risk transactions reveals systematic patterns, including extreme token amounts, repeated sender participation, predictable temporal rhythms, and abnormal behavior under stable liquidity conditions. Together, these patterns strongly suggest automated or algorithmic activity rather than random noise or short-term market volatility.

Importantly, the detected anomalies are not labeled as definitive fraud. Instead, they represent transactions that deviate substantially from learned norms and therefore warrant further investigation. The proposed framework functions as a risk-aware screening mechanism that prioritizes suspicious activity for deeper analysis, rather than as a definitive classification system.

Overall, this study demonstrates that combining complementary unsupervised models with interpretable visual and behavioral analyses provides a robust and scalable approach to anomaly detection in decentralized finance. The framework is fully data-driven, does not rely on labeled examples, and can be adapted to other smart contracts or blockchain protocols. As such, it contributes to ongoing efforts to improve monitoring, transparency, and risk assessment in rapidly evolving blockchain-based financial ecosystems.

GitHub Repository: https://github.com/minaozdogan/Blockchain-Smart-Contract-Transaction-Anomaly-Detection-Project